



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECP 3204: Software Engineering

Semester 01, 2024/2025

PROJECT PROPOSAL

KADA ESERVE SYSTEM

LEMBAGA KEMAJUAN PERTANIAN KEMUBU(KADA)

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1. Introduction

Efficient document management system (DMS) is essential for every organization since it significantly reduce the time required for document handling and optimizing organizational workflows. However, Lembaga Kemajuan Pertanian Kemubu (KADA) still rely on traditional physical document management systems which makes it difficult for them to keep up with current needs. KADA currently aims to improve service delivery and optimize internal processes, there is a pressing need to transition from traditional methods to digital solutions that offer greater efficiency, security, and accessibility.

1.1. Need (N)

As highlighted by KADA's representative Mr. Ahmad Rohailan Bin Hani, the current reliance on paper-based document management at KADA results in inefficient workflows, slow processing and increased risks of document loss. Furthermore, the lack of robust reporting features limits data-driven decision-making which making it harder for them to adapt to rapid changes in information management. There is a critical need for an efficient DMS system that can overcome these limitations, streamline document handling and improve operational efficiency. KADA eServe system aims to address these needs by digitizing and automating KADA's document workflows, significantly enhancing operational efficiency.

1.2. Approach (A)

KADA eServe is designed to improve the efficiency and accuracy of document and workflow management by leveraging a locally hosted database infrastructure. This approach ensures that all data is stored and managed within KADA's internal servers, providing enhanced control and security over sensitive information. The system will streamline processes such as member registration, loan application, and report generation by digitizing workflows and automating repetitive tasks.

Key features include role-based access control for different user and admin, secure authentication mechanisms, automated loan calculations and comprehensive reporting tools to support decision-making. By centralizing all operations within KADA's infrastructure, the system will provide a cost-effective solution with high availability, data integrity, and compliance with local data protection regulations.

1.3. Benefit (B)

KADA eServe system will significantly improve KADA's document management efficiency by reducing manual handling, accelerating document processing, and minimizing the risk of document loss. Users will benefit from streamlined workflows and have a quicker access to documents. Additionally, the system's reporting features will enable KADA to generate a detailed reports for better decision-making that leads to a better productivity and service delivery. Using a local database will allow faster data retrieval and processing while reducing dependency on external internet connectivity. This design particularly suited KADA's operational needs, ensuring uninterrupted service even in areas with limited or unstable internet access. Overall, the system will improve service delivery and support KADA's goals effectively.

1.4. Competitor (C)

Several well-known document management systems, such as Google Drive [1], Microsoft SharePoint[2], and Dropbox[3] offer clouds storage and file-sharing features. However, these platforms do not address the specific needs of government agencies like KADA, particularly in managing member registration, loan applications, and reporting processes.

Google Drive provides secure file storage and sharing but it lacks built-in feature for automating process like loan application tracking or report generation. Similarly, Microsoft SharePoint offers some customization options but can be complex to use, making it less user-friendly for non-technical users. Dropbox is ideal for file storage but does not provide tailored solutions for workflow automation or compliance monitoring tools essential for government operations.

In contrast, KADA eServe is specifically designed to address these gaps. It includes advanced reporting features that generate real-time insights for strategic planning. The system leverages a local database to ensure rapid data retrieval, high performance, and uninterrupted operation, even in areas with limited or unstable internet access.

Moreover, KADA eServe simplifies workflow through an intuitive user interface, reducing the learning curve for users with varying technical expertise. Its seamless compatibility with KADA's existing IT infrastructure ensures smooth transition and integration. By prioritizing functionality, ease of use, and government compliance,

KADA eServe offers a robust and cost-effective solution that surpasses the capabilities of existing platforms while addressing KADA's unique operational needs.

2. Existing Systems

The current systems at KADA present a range of challenges in managing member registration, loan application, and member financial statements. These manual and web-based approaches have significant limitations that affect efficiency, accuracy, and accessibility. Below, we outline the key aspects of the existing manual system and the current websites functionalities before comparing them with others option available.

2.1 Existing Manual System

2.1.1 Member Registration

The existing system uses physical registration forms in which applicants need to fill out these hardcopy forms with their details, name, identification number, salary and contribution amount. After this process, employees are required to cross-check the information for completeness and obtain signatures from managers. The existing system requires a lot of physical storage and space. Filing cabinets and storage rooms take up significant office space.

KOPERASI KAITANGAN KADA KELANTAN BERHAD
PERMOHONAN MENJADI ANGGOTA
MAKLAJUTAN PENCATATAN

DATA PERSANJANG

NAMA: _____
NO. KP: _____
TARAF PERKAHWINAN: _____
ALAMAT RUMAH: _____
POSKOD: _____
NEGERI: _____
NO. ANGGOTA: _____
NO. PF: _____
JAWATAN & GRED: _____
ALAMAT PEJABAT (Tempat Bertugas): _____
POSKOD: _____
NO. TEL/FAX: _____
NO. TEL BIMEIT: _____
NO. TEL RUMAH: _____
Gaji bulanan: _____ RM

DATA PERSEKUTUAN

JANTINA: ☐ LELAKI ☐ PEREMPUAN
AGAMA: ☐ ISLAM ☐ LAIN-LAIN Nyatakan _____
BANGSA: ☐ MELAYU ☐ LAIN-LAIN Nyatakan _____

MAKLAJUTAN PERSEKUTUAN

BIL	HUBUNGAN	NAMA	NO. K/P @ NO. SRT BERSEKUTUAN

* SILA BUKTILAH DENGAN SALAH SATU DOKUMEN YANG BERKUALITI

YURAN DAN SUMBANGAN

JIKA DI TERIMA SEBAGAI ANGGOTA, SAYA BERSETUJU MEMBAYAR YURAN DAN SUMBANGAN BULANAN SEPERTI DI BAWAH :

BIL	PERKARA	RMT
1	FEE MASUK	
2	MODAL SYER *	
3	MODAL YURAN	
4	WANG DEPOSIT ANGGOTA	
5	SUMBANGAN TABUNG KEBAJIKAN (AL-ABRAR)	
6	SIMPANAN TETAP	
7	LAIN-LAIN	

* TAMBAH MODAL SYER HADAPIN KEBANYAKAN RM 200.00 DARI TINGKATAN 1.5 DARI MODAL SYER KOPERASI DARI KEMERULIH SUDAHKAN DALAM TEMPORAL BULAN DARI TINGKATAN KEMERULIH ANGGOTA

Figure 1 : Manual Member Registration Form

2.1.2 Loan Application

The current loan application process at Lembaga Kemajuan Pertanian Kemubu (KADA) relies on a paper-based system. This system requires applicants to complete physical application forms, which are then manually processed by KADA employees. This system has significant challenges in terms of efficiency and accuracy although it has been used for many years.

The image shows two pages of a manual loan application form. The top page is titled 'PERMOHONAN PEMBIAYAAN ANGGOTA' and contains sections for 'BUTIR-BUTIR PEMBIAYAAN' (Loan Details) and 'BUTIR-BUTIR PERIBADI PEMOHON' (Applicant's Personal Details). The bottom page is titled 'BUTIR-BUTIR PENJAMIN' (Guarantor Details) and contains sections for 'PENJAMIN 1' and 'PENJAMIN 2', 'PENDESAHAN MAJLIS' (Majlis Resolution), and 'KEGUNAAN PERAKAT' (Use of Funds). The form includes various fields for names, addresses, contact information, and financial details, as well as checkboxes for different types of loans and guarantors.

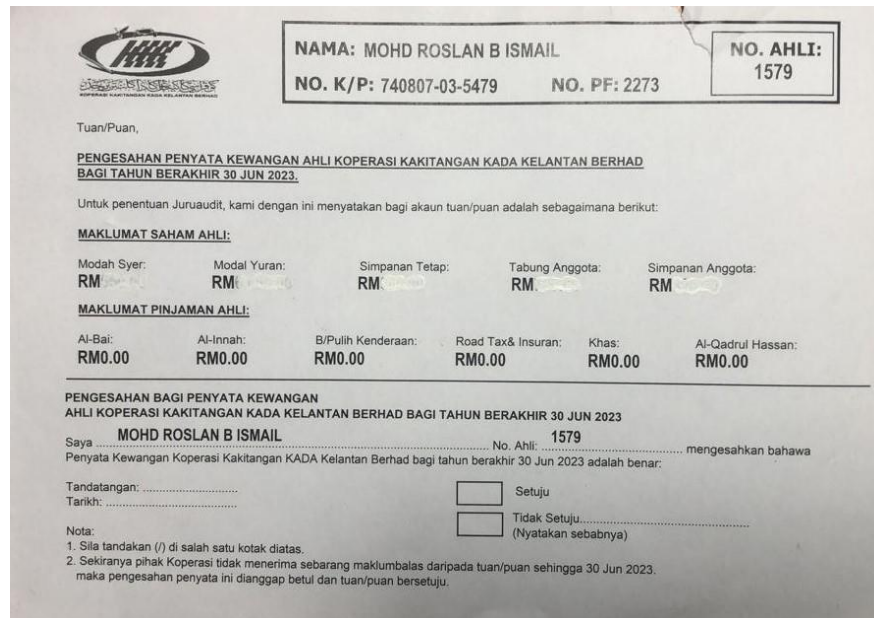
Figure 2 & 3: Manual Loan Application Form

2.1.3 Member Financial System

KADA currently uses a manual system for generating and distributing member financial statements. This involves collecting financial statements manually from various records and compiling it into individual member statements. The member statement requires important details including shares held, loans taken and savings.

Once the applicants fill out the details, they are reviewed by KADA employees to ensure accuracy. After verification, the member statements are processed for the

members. Members will then sign and return the verification to confirm receipt and accuracy. The manual process is time-consuming and prone to errors due to manual data entry. It also imposes a significant burden on KADA's limited human resources and delays members' access to their financial information.



NAMA: MOHD ROSLAN B ISMAIL
NO. K/P: 740807-03-5479
NO. PF: 2273
NO. AHLI: 1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modal Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM 0.00	RM 0.00	RM 0.00	RM 0.00	RM 0.00

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax & Insuran:	Khas:	Al-Qadriul Hassan:
RM 0.00	RM 0.00	RM 0.00	RM 0.00	RM 0.00	RM 0.00

PENGESAHAN BAGI PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya **MOHD ROSLAN B ISMAIL** No. Ahli: **1579** mengesahkan bahawa Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar.

Tandatangan: _____
Tarikh: _____

☐ Setuju
☐ Tidak Setuju (Nyatakan sebabnya)

Nota:

- Sila tandakan (/) di salah satu kotak diatas.
- Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023, maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

Figure 4: Member Financial Statement

2.2 Existing Website

The current website for KADA lacks the functionality found in the manual system, including those that are typically managed manually. These missing features include member registration, loan application, member financial system and automated generation of reports. As a result, KADA's staff and members face significant inefficiencies in managing day-to-day operations and accessing essential services.

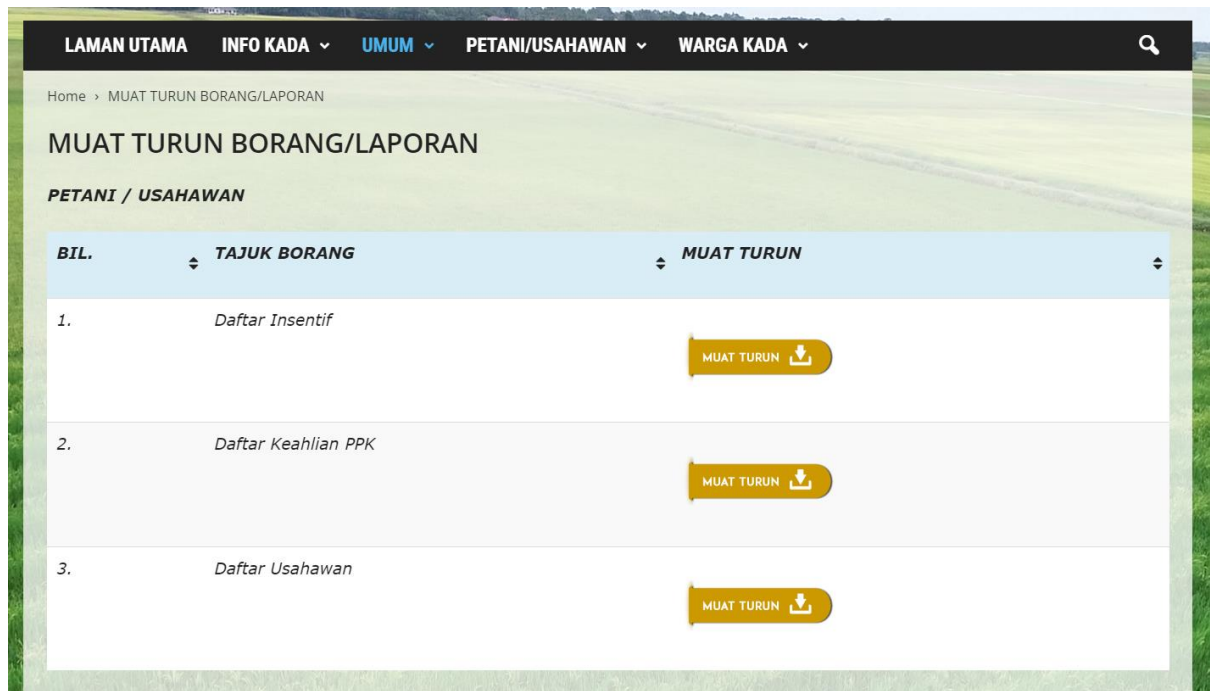


Figure 5: KADA's Forms and Reports Download Page [4]

Based on Figure 5, we can identify that neither the process of member registration nor loan application can be carried out through this website. Instead, users are required to download, print, or request the manual forms from KADA office and fill in all the information manually, which must then be submitted in person. This process is time-consuming, prone to errors, and inconvenient for both members and staff. The lack of a digital system creates bottlenecks in application processing, delays in approvals, and increases the risk of lost or misplaced forms.

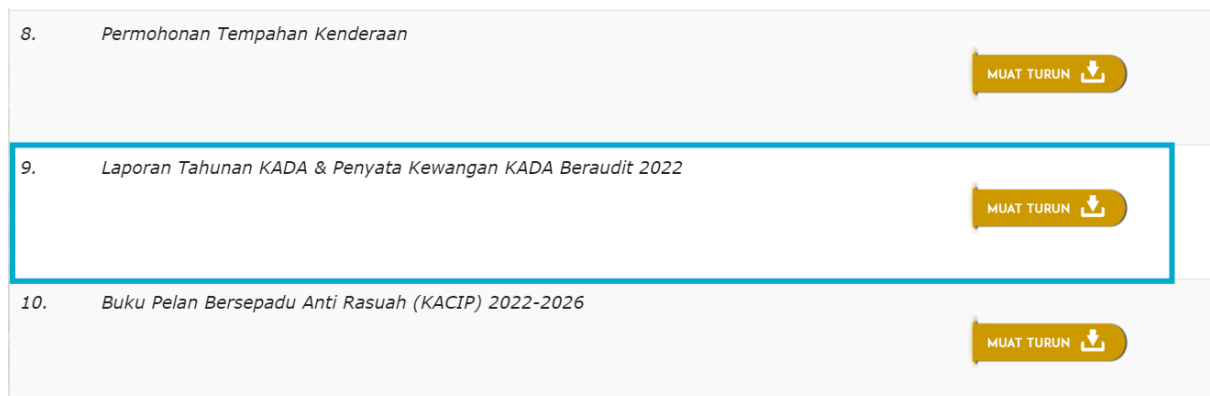


Figure 6: Annual KADA Reports Download Section [4]

Based on Figure 6, we can identify that the existing system does not support automated generation of annual reports. KADA employees are required to manually enter the data to produce an annual

report. This manual approach not only consumes valuable time and resources but also limits the ability to generate real-time insights or respond to urgent data requests effectively.

2.3 Comparative Analysis

To better understand the gaps and limitations in KADA's existing systems, we compared them with other widely used platforms. The following table provides a comparative analysis of key features between KADA and other relevant.

Features	KWSP i-Akaun	Maybank2U	KADA
Digitized Member Registration	Yes	Yes	No
User Authentication for member login	Yes	Yes	No
Automatic generation of annual report	Yes	Yes	No

Table 1: Comparison of existing systems

3. Proposed System

KADA eServe provides a comprehensive solution tailored to KADA's needs, focusing on three core functionalities: **loan application management**, **saving management** and **automated report generation**. Members can easily apply for different types of loans through a streamlined process that includes status tracking and automated calculations of repayment details. Saving management features allow KADA to calculate, update, and manage members' savings based on monthly salary deductions and received dividend details. To support these core functionalities, the system includes several additional features designed to enhance usability, security, and operational efficiency. These include digitized member registration, advanced security mechanisms, email notifications, and user-friendly interface. Each of these features addresses specific aspects of KADA's workflows and combines seamlessly to deliver a cohesive and efficient system. The following sections detail these features and how they contribute to the overall functionality of the system.

3.1 Digitise Member Registration and Data Entry Process

Changing the paper-based system for member registration and loan application process to a digitise system using website. No more paper forms filling process for member registration and loan application and KADA staff are not required to perform manual data entry to register new members. This will mitigate the risk of using paper forms, ensure data security, saving times for checking the documents and most importantly avoid human error.

3.2 Local Database Design and Implementation

The system is built on a locally hosted database, ensuring fast data retrieval and processing while reducing reliance on internet connectivity. This design provides uninterrupted access to the system, even in areas with unstable or limited internet, and ensures full control over sensitive data stored within KADA's infrastructure.

3.3 Security

3.3.1 User Authentication Mechanism

A login page will be created for the system due to security and privacy concerns. After registering an account, the members can login by entering their username and password. Then, they can apply for a loan or check the status of loan applications and payments

made monthly. The system is also able to detect users that are not registered as a member to ensure the security of the system.

3.3.2 OTP Verification

OTP verification is a must for verifying the provided phone number is from the person who register as a member. The OTP verification will be used in member registration process and during change of password for verifying the identification of the user. This is a way to secure our website and data privacy of our users.

3.3.3 Probably destroy section

The current feature is to ensure the website will be logged out while the website is stopped using by our user. This feature is to protect the data privacy of our users and strengthen the security of our website.

3.4 Automated Generation of Report

Currently, KADA is still required to generate reports for the board of directors manually. However, by using our system, the report for each member will be automatically generated after the clerk has updated the loan and dividend details in the system, enabling clerk and management to access real-time insight for better decision-making. Members can also obtain a copy of their individual report via the homepage.

3.5 Clerk's Email Notification

Notification by email will be used to notify the clerk to proceed the steps for checking all the system applications and deciding the result of the application. When there is a new application sent into the system, the system will auto generate an email notifying the clerk to ensure each application will not be ignored.

3.6 User-friendly Layout

3.6.1 Customization layout

By consideration, KADA eServe can adjust the text size that is suitable for different users as each user has different requirement while using website. Otherwise, there are choices for day or night mode for each user based on their preference.

3.6.2 FAQ

Frequently Asked Question section will be provided to solve issues or problems that user faces. This section is needed for them as a guidance while facing problems that are commonly faced by most of the people and to save the time for the server's admin for assisting each issue.

3.6.3 Chatbot

KADA eServe provides a live chatbot that simulates human conversation to help users answer questions or solve problems. This feature is more user-friendly as it can solve the user's problem in a short time.

4. Software Process Modal

4.1 Software Process Model – Proposed System Development

The ideal software process model for internally developing our proposed system is the Agile Development Model, chosen for its flexibility, iterative nature, and focus on user-centric outcomes. Agile emphasizes incremental delivery, allowing delivery of the system in small and manageable parts, and each increment represents a usable portion of the system that is created, tested, and deployed iteratively. It enables early feedback and usability testing as it allows features to be released progressively. For instance, the system could be developed into different functional modules, and referred to as increments such as user registration, member login functionalities, loan applications, and so on.

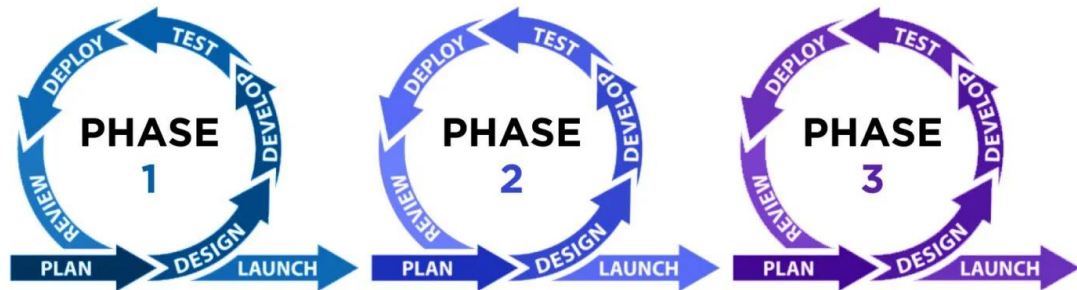
Furthermore, agile's adaptability allows seamless integration of new requirements or modifications as they develop. For example, additional features like enhanced reporting mechanisms or updated security measures can be incorporated without terminating development progress. Agile also emphasizes iterative testing, which ensures high-quality delivery by identifying and addressing potential or existing problems in the development stage, thus minimizing risks and complexities during late integration.

By focusing on incremental and iterative development, agile promotes an organized yet flexible approach to support continuous improvement. Therefore, this model is well-suited for us to develop our system, especially since it allows ongoing discussions with the lecturer and teams to refine and improve the system and guarantee that every component really satisfies the requirements and expectations. Ultimately, agile enables the delivery of a completed and executed product to KADA while adhering to an effective and suitable system development cycle.

4.2 Software Process – Detailed Activities

In agile software development, the activities involved in each phase are not limited to a linear end-to-end process. Unlike traditional development methods, where each phase occurs sequentially, Agile emphasizes continuous, iterative cycles (sprints) in which activities overlap and evolve as the project progresses. Each Sprint consists of repeated processes such as planning, design, development, testing, deployment, and review, all of which are revisited and improved at the beginning of each iteration. This approach ensures

that feedback is continuously incorporated, features are delivered incrementally, and the product can adapt to changing user needs. The following sections outline the detailed activities performed in each Sprint:



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1. Planning Phase

- Product Backlog Creation: The team works with stakeholders to create an initial product backlog, which is a prioritized list of features, user stories, and tasks that need to be done for the product.
- Sprint Planning: At the beginning of each Sprint, the team decides which items from the backlog will be worked on during the Sprint.
- Task Breakdown: Each user story or feature is broken down into manageable tasks, and the time estimates are assigned.

2. Design Phase

- Collaboration with Stakeholders: Continuous input from stakeholders helps shape the design according to emerging requirements.
- System Layout: In the beginning, the design is more high-level and conceptual. As iterations go on, the design evolves.
- Refinement: Each iteration may require improvements to the design as new features are added or adjustments made based on feedback. The team adapts the design to the changing needs of the project.

3. Development Phase

- Coding: The developer starts implementing the tasks defined during the Sprint schedule.

- Daily Stand-ups: The team holds a daily stand-up meeting (or daily Scrum) to discuss progress, obstacles, and next steps.
- Continuous Integration: Developers are constantly integrating their work into the main codebase, ensuring that the code works harmoniously with the contributions of others.

4. Testing Phase

- Continuous Testing: As features are developed, they are continuously tested.
- User Acceptance Testing: UAT can take place at the end of each Sprint, especially if the Sprint delivers a functional feature that can be verified with the product owner or stakeholder.

5. Deployment Phase

- Incremental Release: Each Sprint ends with an incremental release of the software, which is a potentially deployable release.
- Deployment: If the product increment is stable and meets acceptance criteria, then it can be deployed or provided for user feedback. When all system layouts are developed and stable, a final round of testing will take place in a production environment.

6. Review/Maintenance Phase

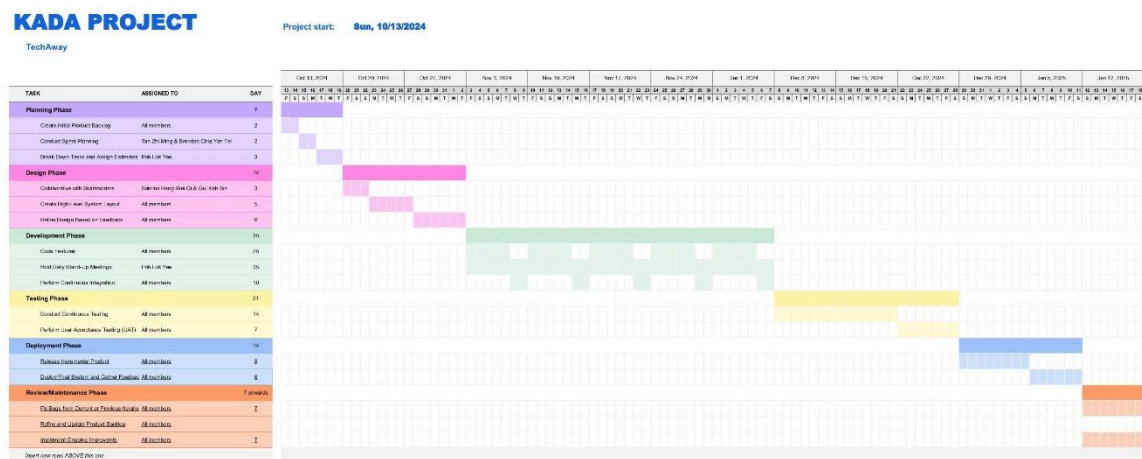
- Bug Fixing: Any bugs found in the current iteration (or previous iterations) will be fixed in the next iteration.
- Backlog Refinement: The product backlog is constantly updated with new tasks, bug fixes, and enhancement ideas based on user feedback or new insights.
- Ongoing Improvement: Based on releases and feedback, the team can adjust priorities and refine features in upcoming iterations.

4.3 Software Process – Deliverables and Duration

Phase	Duration	Person(s) In- Charge	Deliverables
Planning Phase	Week 1	Lecturer, Team Members	Product backlog, sprint plan and task breakdown with time estimates.
Design Phase	Week 2 – 3	Lecturer, Team Members	System layout, evolving and refined designs based on feedback members or lecturer.
Development Phase	Week 4 – 8	Team Members	Completed features, integrated and functional codebase and daily stand-up updates.
Testing Phase	Week 9 – 11	Team Members	Continuous testing results and user acceptance testing (UAT) feedback.
Deployment Phase	Week 12 – 13	Lecturer, Team Members	Incremental software and deployed stable product increments
Review/ Maintenance Phase	Week 14 onwards	KADA, Lecturer, Team Members	Bug fixes, updated product backlog and ongoing improvement adjustments.

5. Project Schedule

Even though our Gantt chart indicates that the project begins with the planning phase this semester, initial planning and design activities were already undertaken during the previous semester. These preliminary efforts have outlined the project's core objectives, defined basic requirements, and drafted initial design concepts. However, this semester marks the commencement of the full implementation process, where we work through the entire software process, including detailed planning, design, development, testing, deployment, and review/maintenance, to bring the project to completion.



6. References

- [1] Google Workspace. Retrieved from <https://workspace.google.com/products/drive/>
- [2] Microsoft SharePoint. Retrieved from <https://www.microsoft.com/en-us/microsoft-365/sharepoint/collaboration>
- [3] Dropbox. Retrieved from <https://www.dropbox.com/business>
- [4] KADA. Retrieved from <http://www.kada.gov.my>